A Quantale-Weakness Route to P $\neq$ NP via CD Evidence Normalization and Gauge-Buffered Locked Ensembles:
A Machine-Checked Formalization Companion

The MeTTapedia formalization effort

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Abstract

This is a living formalization companion to Ben Goertzel's current-best P versus NP manuscript, *A Quantale-Weakness Route to* P $\neq$ NP *via CD Evidence Normalization and Gauge-Buffered Locked Ensembles* (v13, April 2026). That manuscript is a re-architecture of the earlier Switching-by-Weakness route (v12): it discards the "short computation is local" midpoint and replaces it with an *evidence-accounting* one. The proof is a single clash in polytime-capped conditional description length $K^{\text{poly}}(M(Y) \mid Y)$ on an efficiently samplable family of SAT instances Y whose every witness carries the same global readout message $M(Y)$: under P = NP a self-reduction gives $K^{\text{poly}}(M(Y) \mid Y) = O(1)$, while an unconditional lower bound argues $K^{\text{poly}}(M(Y) \mid Y) \geq \eta t$. The lower bound treats computation as an evidence-producing process: predictive advantage becomes constructible-dual (CD) evidence skew, then pairwise distinctions between message-opposite worlds; a normalization theorem forces every target-relevant non-neutral trace leaf to be a safe-buffer or a hidden-gauge observation; an *atomic evidence budget* then caps the total advantage at $o(t)$ across the selected coordinates, which with boundary-law mixing yields product small-success $2^{-\Omega(t)}$ and, by compression-from-success, the linear incompressibility.

We record what the MeTTapedia Lean development has machine-checked of this route, and we are scrupulous about a distinction that a naive axiom audit would blur: the development contains *no* `sorry`, `axiom`, `native_decide`, or placeholder in its proof terms, and every headline reduces to Lean's standard base {`propext`, `Classical.choice`, `Quot.sound`}; but the *open* content of the route is carried *in the types*, as interface-hypothesis fields of structures and as named `Prop` obligations passed as hypotheses. Consequently a clean `#print axioms` on the central clash theorem certifies *the implication, not the conclusion.* We label every result as unconditional, conditional on Ben's named interface hypotheses, or open, and we foreground the places where the formalization *refutes* a stated step (including one of the manuscript's own scaled constructions, proved inadmissible) rather than discharging it. We make no claim about P versus NP, which remains open; the formalization's terminal separation node is open, at 0%. Appendices record the superseded v12 route — whose calibration step this companion's predecessor was devoted to obstructing — and a graveyard of disproven sub-routes, with their history.

Contents

1 Orientation: what this document is

A companion of this kind exists to grow alongside a formalization, not to freeze a single verdict. It has three jobs. First, to distill the *current-best* version of Ben Goertzel's proof — the v13 evidence-accounting route — as a faithful mathematical spine (§2). Second, to record what the Lean development has actually checked of that spine, with an honest ledger separating what holds unconditionally from what holds only modulo the manuscript's interface hypotheses (§§3–5). Third, to keep the history of obstructions and reroutings — the counterexamples the formalization found and how the route moved around them — both inline where they bear on the argument (§6) and, for wholly superseded paths, in appendices (Appendices A–B).

The reader should keep one structural fact in view throughout. The manuscript's proof is abstract-then-concrete: an evidence calculus and a budget theorem are proved *from* a stated interface that a concrete ensemble must satisfy, and the concrete gauge-buffered locked ensemble is the object that must satisfy all of those interface hypotheses at once. The formalization mirrors exactly this shape. Its strongest verified content is therefore of two kinds: genuinely unconditional theorems about the calculus (normalization, bookkeeping, and a battery of counterexamples that kill specific over-strong readings), and conditional *implications* that assemble the interface into the clash. The genuine open mathematics — the concentration and hardness inputs the ensemble must supply — is named precisely and left open.

2 The current-best route (v13)

We follow the manuscript's own notation and numbering. What is preserved from v12 is only the outer skeleton — the polytime-capped weakness object K^{poly}, the Compression-from-Success bridge, and the self-reduction upper bound under P = NP; everything between them is new.

2.1 The K^{poly} upper–lower clash

The cost object (v13 §2.1–2.2, Def. 2.2) is polytime-capped conditional description length $K^{\text{poly}}(x \mid y)$: the length of the shortest program that outputs x from y within a fixed polynomial time bound. The construction builds an efficiently samplable family of public SAT instances Y with a polynomial-time witness relation $R(Y, W)$; every satisfying witness carries a *readout message* $M(W) \in \{0,1\}^{r_t}$, $r_t = \Omega(t)$, $t = \Theta(m)$, and the ensemble enforces the *single-message promise* (v13 §1.1, Prop. 4.10):

$$R(Y, W) = 1 \ \wedge \ R(Y, W') = 1 \implies M(W) = M(W'),$$

so $M(Y)$ is well-defined on the support even though the witness is *not* unique. The proof is the clash of two bounds on $K^{\text{poly}}(M(Y) \mid Y)$:

- **Upper bound, conditional on** P = NP (v13 §11 / Prop. 2.17, Thm. 11.10): if P = NP, the bit-fixing SAT self-reduction finds *some* satisfying W in polynomial time, and the fixed readout returns $M(W) = M(Y)$; this is one fixed constant-length program, so $K^{\text{poly}}(M(Y) \mid Y) = O(1)$ pointwise on the support. Uniqueness of witness is not needed — the single-message promise makes any witness sufficient (Rmk. 11.17).
- **Lower bound, unconditional over the ensemble** (v13 §10 / Thm. 10.9): for every fixed polynomial clock exponent D, after choosing ensemble parameters for that D, there are constants $\eta, \kappa > 0$ with $\Pr[\, K^{\text{poly}}_{(D)}(M(Y) \mid Y) \geq \eta t \,] \geq 1 - 2^{-\kappa t}$.
- **The clash** (v13 Thm. 11.14 / Cor. 11.15): assume P = NP; it supplies a fixed solver exponent D^*. Instantiate the ensemble at D^*. For large t, $\eta^* t > c_{\text{up}}$, yet the same instance must satisfy both $K^{\text{poly}} \geq \eta^* t$ and $K^{\text{poly}} \leq c_{\text{up}}$: contradiction, hence P $\neq$ NP.

The clocked constants (e.g. a safe radius $R_{\text{safe}} = c(D^*) \log m$) are allowed to depend on the alleged solver's exponent D^*; the manuscript defends this (Rmks. 11.16–11.19) on the ground that under P = NP the exponent D^* is one fixed finite constant, instantiated before $m, t \to \infty$.

2.2 The gauge-buffered locked ensemble

The ensemble (v13 §4) replaces v12's masked-and-isolated USAT / Valiant–Vazirani construction. A regioned public geometry G_m (Def. 4.1) partitions sites into neutral, buffer, locked, gauge, and readout regions. The public instance $Y = (Y_{\text{neu}}, Y_{\text{buf}}, Y_{\text{lock}})$ (Def. 4.2) exposes neutral template syntax, the public law of a strictly-positive high-temperature *buffer*, and the public template of the locked global-message layer — and crucially contains *no* raw witness values, no hidden gauge values, and *no* public atom of the form $M_a = b$ (no planted message tag). Hidden sites (Def. 4.3) carry a raw representative x, a hidden *gauge* vector g, and the gauge-invariant quotient z, tied by $z_v = x_v \oplus g_v$.

The verifier (Def. 4.8) conjoins locked-layer acceptance, the gauge equations, buffer acceptance, readout consistency, and the Tseitin/CNF gadgets; it has a uniform poly-size CNF realization F_Y (Thm. 4.13). The gauge group $(\mathbb{F}_2)^V$ acts by $s \cdot (x, g, z, \xi, M) = (x \oplus s,\ g \oplus s,\ z,\ \xi,\ M)$, fixing Y, R, and M (Lemma 4.15); the *hidden-gauge product law* (Hyp. 4.17) keeps unexposed gauge coordinates independent fair bits, giving the guessing bound $\Pr[\hat{g}_J = g_J \mid Y, h] \leq 2^{-|J|}$ (Cor. 4.19). A *legal safe probe* reads few buffer sites far from the locked/gauge/readout regions (Def. 4.25); under buffer contraction / Dobrushin uniqueness (Hyp. 4.27) the soft-buffer max-qSSM bound (Thm. 4.29) makes polynomially many probes leak only $o(1)$ (Cor. 4.30). Thm. 4.31 exports the interface: samplable Y/F_Y; poly-time R; message $M \in \{0,1\}^{r_t}$; single-message promise; gauge action; hidden-gauge product law; atom completeness (Hyp. 4.22); and max-qSSM.

2.3 The evidence-accounting midpoint

The weight of the proof sits on a new midpoint (v13 §1.2):

$$\text{advantage} \implies \text{CD evidence skew} \implies \text{atomic pair-distinctions} \implies \text{safe leakage} + \text{gauge rank}.$$

Its components, in order: **(§3)** for a target bit B_j with opposite fibers, a transcript's positive/negative masses encode predictive advantage as *evidence skew*, and the phase gap $\mathrm{Gap}_j(A_j) = \frac{1}{2}|\Pr[A_j = 1 \mid B_j = 1] - \Pr[A_j = 1 \mid B_j = 0]|$ equals the CD skew (Lemma 3.20). **(§5)** the CD evidence normal form (CD-ENF, Thm. 5.17) leaves only three non-neutral leaf classes — neutral, safe-buffer, hidden-gauge — and gauge faithfulness (Thm. 5.20, Cor. 5.21) forbids using the quotient identity $z_v = x_v \oplus g_v$ as a canonical-gauge shortcut: any target-relevant use normalizes to raw gauge support and is *charged*. This is the technical heart that replaces v12's calibration/involution step. **(§6)** CD trace capture (Thm. 6.15) telescopes an observer's evidence trace, bounding the output phase gap by half the total atomic derivative along the trace: internal computation creates no new pair-distinctions; advantage enters only when a primitive atom separates two previously undistinguished opposite worlds. **(§7.7)** the *Atomic Evidence Budget* (Thm. 7.23) combines these with atom-level bounds — neutral leaves contribute 0, safe leaves are bounded by $Q_{\mathrm{tot}}\, \varepsilon_{\mathrm{step}}(m)$ (max-qSSM), gauge leaves by a gauge-rank entropy rank_G — to give $\sum_{j \in S} \mathrm{Gap}_j(A_j) = o(|S|)$ (Cor. 7.26). **(§8)** boundary-law mixing (Thm. 8.10) makes the pivot-visible surface near-random. **(§9)** combining the budget with mixing yields product small-success $\Pr[A_S = B_S] \leq 2^{-\Omega(t)}$ over the $\Omega(t)$ switched coordinates (Thms. 9.18/9.19) — the average-case unpredictability of the message coordinates that the whole route reduces to. **(§2.2 / App. A.17)** Compression-from-Success turns that exponential small-success into the linear $K^{\mathrm{poly}} \geq \Omega(t)$ lower bound of §10.

A self-imposed correctness check (Rmk. 13.11), "no hidden use of locality," forbids any module from re-importing the very step — v12's Switching-by-Weakness / calibration via the involution T_i — that the v13 pivot was designed to eliminate. This is a candid signal of where the earlier route was fragile; Appendix A records what the formalization found there.

2.4 What the manuscript claims, and what it leaves open

The manuscript claims P $\neq$ NP unconditionally by contradiction (Thm. 1.1). P = NP is assumed only for contradiction, supplying the fixed exponent D^*. Everything else is discharged *from* the ensemble-interface hypotheses of Thm. 4.31, which a concrete construction must satisfy simultaneously; the manuscript is explicit (Rmk. 13.6, milestone M4) that exhibiting one concrete gauge-buffered locked ensemble satisfying all of them at once — single-message promise, SAT realization, atom completeness, gauge faithfulness, hidden-gauge product law, max-qSSM, bounded gauge incidence, *and* boundary-law mixing — is the remaining weight of the proof. Two contraction hypotheses (boundary posterior contraction, Hyp. 8.5; buffer contraction, Hyp. 4.27) are offered explicitly as "prove-or-import." The public-input/neutrality tension is acknowledged as the pivot's pressure point (Rmk. 4.5): the observer may read the entire public Y, and on the single-message support the target bit is a mathematical function of Y, so the claim that full public-input access normalizes to *neutral* context — with any target-relevant use *charged* — is a genuine structural assumption of the ensemble, not a cosmetic restriction.

3 How honesty is encoded in the formalization

Before listing results we fix the reading protocol, because it is the crux of an honest audit of this route.

No placeholders in proof terms. A scan of the entire PNP development finds *no* `axiom`, `sorry`, `admit`, `native_decide`, `opaque`, or `:= True` placeholder in code. Every headline theorem we cite was checked with `#print axioms` and reduces to a subset of Lean's standard base {`propext`, `Classical.choice`, `Quot.sound`} — some to just {`propext`} or {`propext`, `Quot.sound`}.

The open content lives in the types. The route's open steps are carried not as Lean axioms but as *hypotheses*: as assumption fields of a `structure`, and as named `def ...: Prop` obligations passed in as arguments. Consequently, a clean axiom print on a *conditional* theorem certifies that the *implication* is validly proved, and says nothing about whether the hypotheses hold. Throughout this document we therefore tag each result with one of four statuses:

(a) unconditional proved outright (standard axioms, no interface hypotheses);

(b) conditional a proved implication from named interface hypotheses, or a theorem over a `structure` carrying assumption fields — "machine-checked modulo Ben's interface hypotheses";

(c) asserted `Prop` a named `Prop` obligation used only as a hypothesis — no Lean axiom, but analytically open;

(d) open frontier named in doc-comments, not yet a Lean statement.

The route's own itinerary is recorded as data in `Roadmap.lean`, whose terminal node `pnp.global-separation` is `openBackground` at progress 0%: *no final separation theorem is asserted*. Several earlier nodes are marked `blockedByCounterexample` — the formalization *refutes* the stated step rather than overclaiming it; those refutations are themselves unconditional theorems (§5, Appendix B).

4 The evidence spine, formalized (Phases A–D)

The four phases below are all present in the settled core of the formalization. We give the load-bearing statement of each and its status.

4.1 Phase B: the CD evidence normal form (unconditional)

The cleanest genuine win of the spine. `V13CDENF` defines a raw evidence syntax whose leaves are exactly neutral / safe-buffer / hidden-gauge, with *no* observer-output constructor, and proves that the recursive normalizer preserves semantics (v13 §5–6, App. C.7).

```
theorem CDENF_semantics
    (S : EvidenceSemantics Omega Neutral Safe Gauge)
    (E : RawEvidence Neutral Safe Gauge) :
    S.SatNormal (CDENF E) = S.SatRaw E
```

Status (a), unconditional, axioms {`propext`, `Quot.sound`} (it does not even need `Classical.choice`). That the raw syntax has no opaque observer-output atom is a structural design choice that makes the three-leaf classification exhaustive by construction; observer outputs must be expanded through an interface carrying its own semantics theorem (`observerToCDENF_sat`, status (b)).

4.2 Phase A: the finite evidence spine (conditional)

`V13EvidenceSpine` works over a finite rational law (no measure theory), defines the target `Gap` of an observer, and bounds it by half a derivative sum.

```
theorem phaseA_gap_le_half_derivative_sum
    (C : EvidenceSpineBound mu B T g Pair Stage Branch) :
    Gap mu B T g <= (1/2 : Rat) * C.telescoping.derivativeSum
```

Status (b), conditional: the `EvidenceSpineBound` structure carries the capture inequality (`capturesGap`) and the telescoping bound (`sep_le_derivativeSum`) as *assumption fields*; the theorem assembles them. The two-point toy witnesses are unconditional but trivial. This is the §3/§6 telescoping architecture in place, not a concrete-ensemble estimate.

4.3 Phase C: the gauge-buffered locked interface (conditional)

`V13GaugeBufferedLockedInterface` states the ensemble interface as a nine-field structure over the Phase-A/B objects — single-message promise, hidden-gauge product, no public target tags, atom completeness, gauge faithfulness, safe-qSSM, bounded gauge incidence, boundary mixing, admissible histories — and proves three ledger-coherence consequences (e.g. that an admissible history is not target-measurable, and that boundary mixing blocks pivot sufficiency). **Status (b)**: all three are derived *from* the nine assumption fields. A two-point toy jointly inhabits all nine fields (status (a), but trivial): it certifies the interface is *consistent / non-vacuous*, not that any scaled ensemble satisfies it. The manuscript's quantitative estimates (max-qSSM, gauge-rank) appear here only as abstract budget fields.

4.4 Phase D: the upper–lower clash, as a conditional object

`V13ConditionalClash` assembles Phases A–D into the clash *as an implication.* It reuses the checked lower-bound plumbing and isolates the two genuinely open analytic inputs as their own structures: a `CompressionStarSWHardness` field (the average-case witness-bit hardness transfer, $(\star_{\mathrm{SW}})$) and a `SelfReductionUpperHypothesis` field (the P = NP constant-length decoder).

```
theorem v13_upperLowerClash
```

```
    (L : GaugeBufferedLockedInterface ...)
    (P : ParameterRecord L) :
    UpperLowerClash L P
```

Status (b), conditional — and this is the crux of the whole audit. The theorem is a proved implication from the nine-field ledger L plus a `ParameterRecord` that *bundles the two explicitly self-flagged open inputs*. Its own audit tags exactly those two of seven fields `.openInput`. Because the lower chain forces $\eta \le K^{\text{poly}}$ while the upper hypothesis asserts $K^{\text{poly}} < \eta$, a `ParameterRecord` is jointly contradictory — so the theorem encodes the *clash architecture*, not an unconditional falsehood or a separation. Its clean axiom print certifies the implication is valid; it does *not* certify $\mathrm{P} \neq \mathrm{NP}$.

A naming caution. The module `PNPv13AtomicEvidenceBudget` formalizes the *charged-occurrence bookkeeping* that the §7.9 interface export demands — the no-double-counting / no-omission discipline that any such coordinate sum must obey — together with concrete obstruction countermodels. It does *not* prove the manuscript's quantitative Theorem 7.23 ($o(t)$ budget); the safe-leakage and gauge-rank estimates live only as abstract interface fields. The bookkeeping algebra itself is unconditional:

```
theorem coordinateSummedCost_le_atomBudget_of_noDoubleCounting
    (hno : S.NoDoubleCounting) : S.coordinateSummedCost <= S.atomBudget
theorem
    noDoubleCounting_iff_allPositiveCostAtomsCharged_of_coordinateSummedCost_eq_atomBudget

    (heq : S.coordinateSummedCost = S.atomBudget) :
    S.NoDoubleCounting <-> S.AllPositiveCostAtomsCharged
```

and a companion module exhibits a finite surface where a double-counted atom and an omitted atom *cancel*, so aggregate budget equality holds while both structural failures persist — a machine-checked warning that the aggregate budget alone certifies neither side condition.

4.5 The interface as a finite consistency suite

The manuscript itself flags the pressure point (Rmk. 4.5): on the single-message support the target bit $B_j = \ell_j(M(Y))$ is a mathematical function of the full public instance Y, so $B_j = f_j(Y)$. The full public input therefore *cannot* be classified neutral; only a target-blind public skeleton Y_{neu} can, and every target-relevant use of the remaining Y_{chg} must normalize to charged safe-buffer or hidden-gauge evidence. This is the decisive soundness condition of the whole route — and it is the one a formalization can pin down finitely, before any concentration estimate. The development enforces it as a suite of finite consistency lemmas over the interface, each an unconditional theorem:

- message-opposite coupled pairs must agree on a target-blind skeleton, not on the full Y: same-full-Y opposite pairs are empty (single-message), so any statement pairing opposite messages at fixed Y is vacuous;
- a pair-neutral atom family cannot generate the target ($B_j = g(N(\omega))$ with N pair-neutral forbids a nontrivial message-opposite coupling) — the neutral skeleton must be target-blind;
- the full public input carries derivative mass: if $B_j = f_j(Y)$ then every message-opposite coupling puts unit mass on public-bit disagreements, so Y cannot be folded into the neutral class;

- the full σ-field $\sigma(Y)$ is not admissible conditioning for the current coordinate: a C-measurable balanced bit cannot have $\Pr[B_j = 1 \mid C] = \frac{1}{2}$ on positive-probability atoms, so an admissible history may contain only a target-blind projection of Y;

- a boundary summary that is *sufficient* for B_j forces mixing error $\geq \frac{1}{2}$, so the pivot summary must be genuinely non-sufficient;

- no opaque run-atom: the raw evidence syntax has *no* observer-output constructor, so an arbitrary polynomial-time computation cannot be inserted as one uncharged primitive — it must expand through the interface and be normalized by CD-ENF (§4).

These are the finite obligations a concrete ensemble must satisfy, and they are the checkable core of the interface: they say precisely where the scaled Phase-E family failed (§5.2), and they are what a candidate ensemble must pass rather than assume. What they do *not* supply are the quantitative estimates — max-qSSM, gauge-rank entropy, boundary-law mixing — which remain interface hypotheses (§7).

5 The concrete instantiation: Phase E, the observer ladder, and a refuted family

The manuscript's remaining weight is the concrete ensemble. The formalization's Phase-E work is where it is most valuably *negative*, and the honesty protocol of §3 earns its keep.

5.1 A concrete smoke test that works

`V13PhaseEConcrete` builds a genuine four-world locked unit-CNF instance with an isolation row, proves it has a unique witness, and exhibits a real nonzero target gap Gap $= \frac{1}{2}$ (a finite `norm_num` fact over the four worlds). All nine interface obligations are discharged one by one on this instance — **status (a)**, but explicitly a smoke test: the caveats note that scaled max-qSSM and scaled boundary-law mixing remain separate obligations.

5.2 A scaled family, proved inadmissible

`V13PhaseEScaled` attempts a parameterized hard family — and proves its own construction broken. The public input sets the locked target tag equal to the target itself, so a single public coordinate determines the target:

```
theorem phaseEScaled_familyInadmissible_publicTargetTag (m k : Nat) :
    exists coordinate : PhaseEScaledPublic m k -> Bool,
      forall omega,
        coordinate (phaseEScaledPublicInput omega) = phaseEScaledTarget omega
```

Status (a), unconditional (axioms {`propext`}). This violates the manuscript's own "no public target tags" requirement (Hyp. 4.21), so the family's obligation map re-tags all nine fields *inadmissible*. The lesson, carried forward as a design rule: a candidate ensemble must be pinned by theorem to carry no uncharged target-bearing public atom, or it silently prints the answer.

5.3 The observer ladder

`V13ObserverLadder` climbs the scaled family and records, at each rung, a machine-checked fact — most of them negative. A $q = 1$ observer that reads the public target-lock wins with probability 1 (`phaseEScaled_targetLockObserver_success_eq_one`); target-blind juntas and parities sit at exactly $\frac{1}{2}$; the full random-switching lift is *pinned, not proved*, as `PhaseEScaledSwitchingStatementNeeded` (status (c)); and hardness is proved *equivalent* to the open ($\star_{\mathrm{SW}}$) half-bound:

```
theorem phaseEScaled_payloadDomination_iff_starSW ... :
    CoarseDominationAll payloadSummary target decoderSuccess slack
      <-> StarSW decoderSuccess slack
```

This equivalence is the honest localization: on this family, "a local rule dominates every decoder" is not weaker than the average-case hardness one is trying to prove — it is the same statement.

5.4 The linear replacement route, and a lesson from formalization

After the scaled family was proved inadmissible, a concrete linear-algebra testbed was opened (the Roadmap ties it under `v13RealRungOne*` / `v13ObserverLadder`): public instance (A, Ax) over $\mathbb{F}_2^m$ with A a certified invertible linear map, and a family of row-query observers on it. Two success bounds hold **unconditionally** on this route (axioms {`propext`, `Classical.choice`, `Quot.sound`}), using a crude one-time union-bound constant:

```
-- sequential q-row observers:
--   uniformSequentialQRowSuccess observer i0 <= 1/2 + 4*(2^q - 1)/2^m
-- no-target bit-junta observers:
--   NoTargetBitJuntaSuccess i0 observer <= 1/2 + 4*2^j/2^m
```

A fixed-map parity observer's success obeys an exact trichotomy — 1, 0, or exactly $\frac{1}{2}$ (status (a)). The *sharp* $1/2 + 2^q/2^m$ bound for general causal row queries is open for $q \geq 2$ (a fan of asserted `Prop` counting obligations, status (c)/(d)); the $q \leq 1$ cases are discharged, and a machine-checked *negative* shows the sharp bound is *false* for a target-row-reading observer at $m = 2$, which is exactly why the construction restricts to the no-target event.

This route is also where the formalization taught us something about *stating* events. The natural step-indexed "first coset hit at step t" event, used as a counting target, turned out to be *false as stated*: it carried no "no earlier hit" clause, so an observer that re-reads a row over a prefix that already generates the target makes *every* world in the prefix class hit. The refutation is an explicit constructed observer, not an abstract objection:

```
def rowShearGeneratedRereadObserver (i0 : Fin m) (hm : 1 < m) :
    SequentialRowObserver m 3 where
  chooseRow transcript :=
    match transcript.length with
    | 0 => i0
    | 1 => finSpare i0 hm
    | _ + 2 => i0
  decideFromTranscript := fun _ => 0
-- ...ConditionedCountingBound_not_of_generated_reread_prefix_gap :
--   not (ConditionedCountingBound i0 observer)     -- status (a), a refutation
```

The repair restores the missing minimality: the corrected *true-first* event adds "no coset hit at any earlier step," and with that clause the conditioned, packing, and counting bounds all close *unconditionally.* The crown counting lemma is a conditioned replay of a two-row shear embedding applied at the hit row; because the shear fixes the already-read rows, the conditioning is free and the constants land exactly. The refuted step-indexed surfaces are kept in the development, documented against their refuting theorems, because a refutation of a plausible statement is itself a finding. With the corrected event, these bounds assemble into an unconditional closure of "rung one" on both samplers — a unified success $\leq \frac{1}{2}+\varepsilon$ headline with an explicit rung ledger — and the route's separation endpoint follows from an $(\star_{\mathrm{SW}})$ hardness input by an implication whose axiom print is *empty*: pure plumbing, asserting no separation. (This is the concrete-instantiation frontier; the evidence spine of §4 is the more settled core.)

5.5 The linear gauge-CNF regression carrier and the faithful M4 specification

The current `V15RealLinearCNFEnsemble` and `V15SmallSingleMessageCNFEnsemble` are frozen regression carriers, not the manuscript's M4 ensemble. The general linear carrier unit-locks the decoded witness coordinates and has one free Boolean gauge coordinate. It is useful for testing generic SAT/readout, gauge-action, and bit-fixing machinery, but it has no regioned neutral/buffer-/locked/gauge/readout geometry and cannot supply the growing gauge entropy or buffer law used by the analytic argument.

Its principal manuscript-facing result is negative. When the full concrete Appendix-I formula syntax is public, that syntax determines the target and the readout. Accordingly, `v13RealLinearNoTargetRowsGaugeCNFAppendixICNFFormula_noPublicTargetTags_obstruction` and `v13RealLinearNoTargetRowsGaugeCNFAppendixICNFFormula_noPublicReadoutTags_obstruction` prove that the no-public-tag disciplines *fail* on this carrier. A theorem for a deliberately smaller neutral skeleton does not repair the full-syntax failure. Likewise, the carrier's Appendix-D-shaped read predicate asserts `message = targetBit`; its resulting `PublicMessageInvariant` is premise-baking and is not evidence that D.8 was constructed from bounded-arity lock/readout constraints.

The manuscript-facing work is now a statement-only layer on one common size- and clock-indexed ensemble. `V13FaithfulQuantitativeFrontiers` states the conditional hidden-gauge product law, a nontrivial indexed finite gauge action, full-public-syntax coverage, max-qSSM, finite-set bounded incidence together with the simultaneous Atomic Evidence Budget, conditional boundary mixing, the fixed-clock Theorem-10.9 probability bound, and the Section-11 upper side on the same object at D^*. `PNPv13FaithfulM4Spec` states the five-region Appendix-D/I type layer, the $x/g/z$ quotient equations and uniform gauge lift, lock/readout and buffer predicates, legal probes, sampler support, compatible couplings, and uniform CNF compilation. D.8, D.30, D.33, and D.48 remain exact named construction hypotheses.

No inhabitant of this faithful M4 construction record is exhibited here. The four analytic packages — max-qSSM; bounded incidence plus the Atomic Evidence Budget; conditional boundary mixing; and fixed-clock global message incompressibility — remain explicit open mathematics on that same prospective object. The terminal separation node remains at 0%.

The development subsequently sealed the architecture's honest completion point on a faithful regioned scaffold: a single same-object conditional endpoint theorem at a fixed clock exponent D^*, deriving the upper–lower clash from exactly the enumerated named inputs — the mechanical construction record (including the open D.8, conditional gauge-product package, D.33 residual, and D.48 fields), the quantitative analytic frontiers, and the Section-11 same-object upper premise at that clock. An axiom-free machine-readable hypothesis audit exhaustively classifies all eighteen premise rows (seven constructed scaffold fields, two conditional gauge-package fields, three named construction hypotheses, five analytic frontier inputs, one Section-11 upper); an unnamed leakage-equality premise discovered by the audit was removed rather than carried silently. On the scaffold, D.30 full-syntax no-public-tag coverage and a positive-rank free gauge action are proved outright, while D.8 itself is additionally blocked upstream: the manuscript's Appendix D supplies no concrete bounded-arity locked-CSP clause family, so locked-message rigidity is an explicit hypothesis of the manuscript, not a consequence of a stated construction. The endpoint is the complete v13 separation architecture as one implication from exactly the named open content; it asserts no separation.

6 Obstructions and reroutings

The move from v12 to v13 was not cosmetic; it routed around a specific, now machine-checked obstruction. We record the live reroutings here; the superseded v12 route in full is Appendix A.

The calibration defect, and the kernel-flip repair. v12's calibration (Lemma 4.8 / App. A.17) needed, at a fixed local input, exchangeability of the witness bit under a promise-preserving involution T_i. `tiInputMap` sends the third coordinate $b \mapsto b \oplus a_i$, so it fixes the full input *iff* the column a_i is zero (`tiInputMap_eq_self_iff_zeroColumn`), a vanishing $m^{-\Theta(1)}$ fraction of columns (`not_calibrationFullInputPreservation_on_nonzeroColumns`, status (a)). The repair that strengthens the step is a *kernel-flip* involution: flip the witness by a kernel vector $w \in \ker(A)$ with $w_i = 1$; this toggles the target bit while fixing b (since $Aw = 0$) and the rest of the local statistic, giving *exact* neutrality, expressed as a cardinality-preserving pairing of the two witness-bit fibers:

```
theorem kernelFlip_exactNeutrality_card
    (hw : IsKernelVector A w) (hwi : w i = 1) :
    Fintype.card (KernelBitFiber A b i 0)
      = Fintype.card (KernelBitFiber A b i 1)
```

Exact neutrality removes the broken calibration lemma outright and makes the associated sparsification stack unnecessary.

The No-Threading impossibility. Any hope of *enriching* a single local statistic to be simultaneously neutral and sufficient is blocked, and for a logical rather than analytic reason: the target bit is a deterministic function of the instance, so a sufficient statistic pins it to $\{0, 1\}$, which neutrality forbids from equalling $\frac{1}{2}$.

```
theorem noThreadingLemma (phi : Phi) :
    not (Neutral u x /\ Sufficient u x)
```

Status (a). Because the obstruction is logical, no concentration or log-Sobolev inequality can ever supply the missing domination; this is the structural reason the hardness input $(\star_{\mathrm{SW}})$ cannot be made a free lemma, and part of why v13 abandons the locality midpoint entirely.

The steelman conditional. The repaired abstract implication consumes two premise packages: kernel-flip neutrality plus the open $(\star_{\mathrm{SW}})$ hardness input, and a `PNPConditionalFramework` supplying pivot independence, compression from success, and the self-reduction clash.

```
theorem pnp_steelman_starSW_frameworkBundle_conditional
```

```
    (hKernel : KernelFlipNeutrality F)
    (hSW : StarSWAverageCaseWitnessBitHardness F) :
    F.pNeNPClaim
```

`pnpSteelmanFrameworkPremiseBundleAudit` lists the three non-($\star_{\mathrm{SW}}$) framework premises. **Status (b), conditional**: the composition theorem is axiom-free, but it proves none of those premises and asserts no unconditional separation. A future concrete theorem must establish every premise on the same faithful M4 ensemble. (($\star_{\mathrm{SW}}$) itself is not weaker than the target: it is contradicted in the P = NP regime and is equivalent, on the ensemble, to the separation — which is why No-Threading forbids deriving it for free.)

7 Status, scope, and the open frontier

Machine-checked unconditionally. The syntactic CD-ENF semantics-preservation theorem (not the open real-execution primitive expansion) (§4); the atomic-evidence *bookkeeping* algebra and its cancellation/exactness countermodels; non-vacuity of the nine-field interface; the Phase-E *negatives* (the scaled family is inadmissible; the observer ladder stops at a $q = 1$ target-lock counterexample; hardness $\Leftrightarrow$ ($\star_{\mathrm{SW}}$)); the Roadmap-node refutations of stated readout and calibration steps; and, on the linear route, the two crude-constant success bounds, the parity trichotomy, the public-surface certificate, the $q \leq 1$ causal cases, and the true-first counting close.

Machine-checked only modulo explicit premise bundles. The legacy Phase-A theorem is a conditional certificate eliminator; the faithful Phase-A record still requires normalized exact phase marginals, actual-trace capture, and the refinement comparison. The Phase-C coherence lemmas use the legacy thin interface. The **Phase-D upper–lower clash** is an implication from the full `ParameterRecord`, which supplies the phase budget, lower framework, kernel neutrality, ($\star_{\mathrm{SW}}$) transfer, and the incompatible Section-11 upper premise. These are reduction skeletons, not a separation.

The no-target sequential result is *not* conditional on the former pinned counting proposition. `V13RealLinear_rungLedger` records unconditional half-plus-epsilon bounds for both the adjusted no-target sequential sampler and the bit-junta sampler, together with unconditional Crux-2; only the ($\star_{\mathrm{SW}}$)–framework endgame remains conditional. Separately, the faithful quantitative frontiers and Appendix-D/I M4 layer are open statement/specification records: no concrete M4 inhabitant currently transfers the mechanical interface or the four analytic inputs.

Open. The sharp $2^q/2^m$ causal counting bounds for $q \geq 2$; the random-switching lift `PhaseEScaledSwitchingStatementNeeded`; the ($\star_{\mathrm{SW}}$) average-case hardness input; the two "prove-or-import" contraction hypotheses (boundary posterior contraction, buffer contraction); and — per the Roadmap's terminal node — the entire global complexity-class interface and the final separation, at 0%.

P **versus** NP **remains open.** We make no claim in either direction. The value of the formalization is the verified plumbing and the exact localization of the open inputs, together with a body of machine-checked counterexamples that keep specific over-strong readings of the route honest. The concrete gauge-buffered locked ensemble that would simultaneously satisfy every interface hypothesis — the manuscript's own stated remaining obligation — is not exhibited here.

Provenance

The manuscript distilled in §2 is Ben Goertzel, *A Quantale-Weakness Route to* P $\neq$ NP *via CD Evidence Normalization and Gauge-Buffered Locked Ensembles* (v13, 20 April 2026), which supersedes the earlier P $\neq$ NP*: A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity* (v12, 9 October 2025) discussed in Appendix A. The formalization is work of the MeTTapedia formalization effort. The v13 evidence spine (Phases A–D), Phase-E instantiation, and observer ladder are the modules `V13EvidenceSpine`, `V13CDENF`, `V13GaugeBufferedLockedInterface`, `PNPv13AtomicEvidenceBudget`, `V13ConditionalClash`, `V13PhaseEConcrete`, `V13PhaseEScaled`, and `V13ObserverLadder`, with the itinerary in `Roadmap.lean`; the v12 obstruction is `PostSwitchInputObstruction`; the steelman conditional and its inputs are `PNPSteelmanConditional`, `KernelFlipInvolution`, and `NoThreadingLemma`; the linear replacement route is the `V13RealRungOne* / V13RealRungTwoBitJunta` family; the frozen legacy staging spine and linear CNF regression/obstruction carrier are `V15RealEnsembleSpine` and `V15RealLinearCNFEnsemble`. The faithful manuscript-facing statement layers are `V13FaithfulQuantitativeFrontiers` and `PNPv13FaithfulM4Spec`; they specify, but do not construct, M4. Lean listings are ASCII transliterations of the source declarations (the sources are normative); every headline was checked with `#print axioms`. Per-declaration source links are omitted pending publication of the development tree.

A The superseded v12 route: a machine-checked obstruction in the calibration step

The v12 manuscript separated P from NP by a Switching-by-Weakness normal form (Thm. 4.2) that reduces every short global decoder, after a short wrapper, to an $O(\log m)$-local per-block rule on the post-switch input $u_i = (z, a_i, b)$ — z a sign-invariant feature, (a_i, b) Valiant–Vazirani labels with $a_i = Ae_i$ — on a masked-random Unique-SAT ensemble; neutrality (Thm. 5.1) and template sparsification (Thm. 5.10) pin each local rule near $\frac{1}{2}$; independence across $\Theta(m)$ blocks gives per-program success $2^{-\Omega(t)}$ and hence $K^{\text{poly}} \geq \eta t$, clashing with a constant self-reduction upper bound under P = NP. The load-bearing step was the *calibration*/success-domination link (Lemma 4.8; App. A.5, Lemma A.17), which derived exchangeability of the witness bit at a fixed u from a promise-preserving involution T_i.

The formalization (`PostSwitchInputObstruction`) found that T_i acts by $b \mapsto b \oplus a_i$: it preserves the reduced projection (z, a_i) always (`invariantProjection_tiInputMap`) but the full input (z, a_i, b) *iff* $a_i = 0$ (`tiInputMap_eq_self_iff_zeroColumn`), so the fixed-u exchangeability holds only on a vanishing $2^{-k} = m^{-\Theta(1)}$ fraction of columns. A finite two-point success-domination counterexample (`finite_successDomination_counterexample`, axioms {`propext`, `Quot.sound`}) shows a u-measurable comparator cannot dominate a global decoder merely by sharing the local input. The natural repair — coarsening to the T_i-invariant projection (z, a_i) — was proved *circular*: on that neutral interface, success domination is equivalent to the average-case hardness it must prove (it renames the target). These are the findings the v13 re-architecture routes around by discarding the locality midpoint; the kernel-flip repair and the No-Threading impossibility of §6 are the constructive residue that carries into the v13 steelman conditional.

B A graveyard of disproven sub-routes

Each entry is a route or reading that was tried and closed; several are themselves machine-checked theorems (that a stated property *fails*).

Fixed-input calibration via T_i (v12). Dead as stated: the involution moves b off the fixed fiber on every nonzero column (Appendix A). Superseded by the kernel-flip involution, which gives exact neutrality.

Coarse / reduced-projection repair. Circular: on a neutral interface, "every decoder is dominated by the best local rule" is equivalent to ($\star_{\mathrm{SW}}$) (a two-point witness makes coarse-domination demonstrably fail). A closed door, not a weakening.

Analytic-concentration closure. Retired by No-Threading: the obstruction to a neutral-and-sufficient statistic is logical, so no hypercontractivity or log-Sobolev device can close it.

Per-map capacity counting. Arithmetically dead: a single adversarial map can host a constant fraction of hitting worlds, so the required smallness lives in the map *ensemble*, not in any one map; the closure runs through a per-prefix partition and a *conditioned* one-row density instead.

Surrogate-ERM "distillation" of domination. Relativizes, hence (by Baker–Gill–Solovay) cannot establish a class-separating switching step; symmetrization projects onto an $\approx m$-bit invariant algebra, far richer than the $O(\log m)$-bit local statistic where neutrality lives.

Håstad-style p-random bit restriction. Wrong tool for this family: the public bits are parities of hidden bits at a fixed witness, so random bit-fixing leaves live parities and the switching-lemma hypothesis fails; rerouted to gauge-compensation.

The naive $2^q/2^m$ row-observer constant. Refuted at $m = 2$ by a target-row observer; repaired to the honest $\frac{1}{2} + 4(2^q - 1)/2^m$ crude-constant bound.

The scaled Phase-E hard family. Proved *inadmissible* — a single public coordinate determines the target (§5.2); the family prints the answer. Retained as a design rule: pin no-public-target-tag by theorem, or the ensemble is void.

Token-ledger false starts. An early formalization tracked the argument as status *tokens* with no representation of the actual SAT / Kolmogorov / measure objects; disavowed as proof content and replaced by the finite evidence calculus grounded in the manuscript's objects. Lesson carried forward: never let a family be chosen that carries the answer as a public tag, and demand admissibility by theorem.